

Dimensions of quantum codes from Cayley graphs over $\mathbb{F}_2^r$

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Abstract

The adjacency matrix of a Cayley graph of even degree over the additive group $\mathbb{F}_2^r$ can be used to define a quantum code. Such matrices can also be used to define storage codes of unusually high rate. We characterise the sets of generators of the Cayley graph that give rise to quantum codes of non-zero dimension. We also give formulas for the dimension of the codes in a number of instances.

Key Words : Quantum codes, storage codes, group algebra codes.

1 Introduction

Context

We are interested in the family of quantum codes studied in [CDZ13] and that are defined from a Cayley graph over the additive group $\mathbb{F}_2^r$. Specifically, if S is a set of elements of $\mathbb{F}_2^r$, then it defines a Cayley graph Γ where the vertex set of Γ is $\mathbb{F}_2^r$ and two vertices v, w are connected if $v + w \in S$. It is straightforward to check that if $|S|$ is even, then the adjacency matrix A of Γ satisfies $A^2 = 0$, viewed as a matrix over $\mathbb{F}_2$, and therefore defines a quantum CSS code $\mathcal{C}$ on Γ , whose X and Z sets of generators are both given by the rows of A . The length of the quantum code is 2^r and its dimension equals $2^r - 2\text{rk}(A)$. If the set S is small compared to 2^r , then the matrix A is sparse and defines a quantum LDPC code. In [CDZ13] a particular set of generators is studied that yields codes of parameters $[[2^r, 2^{r/2}, 2^{r/2}]]$ for even r , with some additional interesting properties. A question that was left

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very much open however, is that of characterising the sets S for which the quantum code has non-zero dimension, equivalently the sets S for which the matrix A has rank less than 2^{r-1} .

This question stayed dormant for some years until it came up again, somewhat unexpectedly, in the context of storage codes on graphs [BZ22]. A storage code [Maz15] on a graph Γ is a (classical) linear code whose word symbols are indexed by the vertices of Γ and with the property that any erased symbol on a vertex v can be recovered from the symbols hosted by the neighbouring vertices. The main problem considered in [BZ22] was whether there exist families of storage codes on triangle-free graphs with rate tending to 1. Good candidates to solve this problem turned out to be Cayley graphs on $\mathbb{F}_2^r$ for which the corresponding storage codes are generated by their adjacency matrices A (with the understanding that the set S contains the zero element). The dimension of the storage code is simply the complement of the rank of the matrix A . See [BZ22] for discussions on the relevance of the determination of the rank of A to other problems. The problem of whether storage codes on triangle-free Cayley graphs over $\mathbb{F}_2^r$ can achieve asymptotic rate 1 was not solved in [BZ22], but it was solved in subsequent works [BSY24, HX23, DHWX24].

Let us consider the matrix H whose columns are made up of the columns of the generating set S of the Cayley graph. The paper [BZ22] highlighted that the rowspace of H plays an important role in determining the rank of A . Specifically, the following theorem was proved.

Theorem 1. *Let $\Gamma = \text{Cay}(\mathbb{F}_2^r, S)$ be a Cayley graph over $\mathbb{F}_2^r$ with set of generators S . Let H be the $r \times n$ matrix whose columns are made up of the elements of S , so that $n = |S|$, and let C be a code of length n with parity-check matrix H . Finally, let $\mathcal{C}$ be the quantum code on Γ . Denoting by $R(\mathcal{C}) = \dim \mathcal{C} / 2^r$ the rate of $\mathcal{C}$, we have that:*

- (a) *If $R(\mathcal{C}) > 0$, then n is even and all the rows of H have even weight;*
- (b) *If $R(\mathcal{C}) > (1 - 2^{1-h}), h = 2, 3, \dots$, then $(C^\perp)^{*^{(h-1)}} \subset C$.*

In condition (b) above, the notation $(C^\perp)^{*^{(h-1)}}$ stands for the $(h-1)$ -th Schur power of C with itself. Recall that the Schur product of two codes C_0 and C_1 is the linear span of all coordinate-wise products of a codeword of C_0 with a codeword of C_1 .

Remark 1. *In the context of storage codes, it is required that S contains the 0 element. For quantum coding applications, the set S may or may not contain the 0 element.*

Contributions

Theorem 1 goes some way towards characterising Cayley graphs Γ for which the associated quantum code has non-zero dimension: indeed, it proves that

(a) is a necessary condition for the dimension to be non-zero. In the present work, we prove that condition (a) is also sufficient, in other words it gives the long-awaited characterisation.

We will adopt a group-algebra point of view: as was stated in [CDZ13], the adjacency matrix A represents multiplication by an element f of the group algebra $\mathcal{A}$ of the additive group $\mathbb{F}_2^r$. The algebra $\mathcal{A}$ can be constructed as $\mathbb{F}_2[y_1, \dots, y_r]/I$ where I is the ideal generated by the polynomials $y_i^2 + 1, i = 1, \dots, r$. Computing the rank of A translates therefore into computing the dimension over $\mathbb{F}_2$ of the principal ideal $\langle f \rangle$ of $\mathcal{A}$. This point of view was developed significantly in [HX23]. In the present work we make extensive use of an added twist consisting of applying the change of variables $x_i = y_i + 1$. When viewing elements of $\mathcal{A}$ as polynomials f in the variables x_i , condition (a) is reformulated as f not containing monomials of degree less than 2, and the more general condition (b) is equivalent to f not containing monomials of degree less than h . We have found this formulation helpful in our analysis.

One could naively ask whether the necessary conditions (b) are also sufficient. We will show that this is not the case in a rather strong sense. Indeed, there exist asymptotic families of Cayley graphs Γ such that $(C^\perp)^{*(h-1)} \subset C$ in condition (b) holds for arbitrarily large constant values of h and for which the limsup of the rate of the associated quantum codes equals zero. This fact will be a consequence of the formula presented in Section 3 that studies the behaviour of group algebra elements that are sum of polynomials in disjoint sets of variables. Finally, Section 4 is a study of polynomials f containing a quadratic form of maximum algebraic rank. In fact, when f is itself a quadratic form, we have a detailed description of the associated quantum code together with a formula for its rate.

Not all quantum codes constructed in this way will be interesting of course, one would need in particular to investigate their minimum distances: in the present paper we focus solely on the dimension issue, which we have found to be a rich and intriguing subject in itself and, as discussed above, is amenable to a number of applications outside quantum coding.

2 Characterisation of non-zero quantum codes

To rephrase the setting in the group algebra guise, the ambient space of the investigated codes is the group algebra $\mathbb{F}_2[G]$, where G is the additive group $\mathbb{F}_2^r$ written multiplicatively. Alternatively, $\mathbb{F}_2[G]$ can be seen as the quotient of the polynomial ring $\mathbb{F}_2[y_1, \dots, y_r]$ by the ideal $\langle y_i^2 + 1, i = 1, \dots, r \rangle$, identifying the variable y_i with the i -th generator of G . In this view, we consider the element $f_S = \sum_{v \in S} y^v$ in $\mathbb{F}_2[G]$, the sum of monomials arising from elements of the set S of generators of the Cayley graph. The matrix multiplication by $A(\Gamma)$ coincides with the multiplication by the polynomial f_S in the polynomial ring. Moreover, the even parity of $|S|$ implies that the

quantum CSS code $\mathcal{C}$ thus defined is non trivial and in particular it has the same dimension as the quotient space $\ker f_S / \text{Im } f_S$.

The conditions in Theorem 1 get reformulated as:

(a) if $R(\mathcal{C}) > 0$, then the support of f_S is even and all variables appear in an even number of supporting monomials of f_S ;

(b) if $R(\mathcal{C}) > 1 - 2^{1-h}$, $h = 2, 3, \dots$, then every set of h different variables appears in an even number of monomials in the support of f_S .

To handle the parity conditions more easily, we perform a change of variables ψ in the group ring $\mathbb{F}_2[G]$ setting

$$x_i = y_i + 1.$$

The object of the investigation becomes $\psi(f_S) = f \in \mathbb{F}_2[\bar{x}_1, \dots, \bar{x}_r]$, where $\bar{x}_i$ denotes the image of x_i in the quotient $\mathbb{F}_2[x_1, \dots, x_r] / \langle x_i^2, i = 1, \dots, r \rangle$. It will be convenient to view the group algebra directly as the quotient space $\mathbb{F}_2[x_1, \dots, x_r] / \langle x_i^2, i = 1, \dots, r \rangle$, which we shall denote by $\mathcal{A}_r$ or simply $\mathcal{A}$ when the number of variables is clear enough. To lighten notation we shall write the functions $\bar{x}_i$ simply as x_i , as they will always be considered in the ring $\mathcal{A}$ rather than the original polynomial space $\mathbb{F}_2[x_1, \dots, x_r]$.

After this change of variables, Theorem 1 can be expressed more simply, as the conditions on the parity of the number of appearances of variables in monomials translate into the vanishing of low degree terms in the polynomial f .

Theorem 1 (Second version). *Let $\Gamma = \text{Cay}(\mathbb{F}_2^r, S)$ be a Cayley graph over $\mathbb{F}_2^r$ with set of generators S . Let $f_S = \sum_{v \in S} y^v$ be the element in $\mathbb{F}_2[G]$ that corresponds to the multiplication by $A = A(\Gamma)$ and let $f = \psi(f_S)$. Let $\mathcal{C}$ be the quantum code on Γ . Then*

(a) *if $R(\mathcal{C}) = R(\ker f / \text{Im } f) > 0$, then the degree 1 term of f is zero;*

(b) *if $R(\mathcal{C}) = R(\ker f / \text{Im } f) > 1 - 2^{1-h}$, $h = 2, 3, \dots$, then all terms of degree at most h of f are zero.*

It is convenient to define the *valuation* of a non-zero polynomial $0 \neq f \in \mathcal{A}$ as

$$v(f) = \min\{\deg(m) : m \in \text{supp}(f)\},$$

where $\text{supp}(f)$ is the support of f , namely the set of monomials appearing in f with non zero coefficient. We also set $v(0) = \infty$.

Remark 2. *The valuation in this sense does not satisfy the standard properties of non-archimedean valuations, since $v(fg)$ may be strictly greater than $v(f) + v(g)$. Nevertheless, the triangle inequality $v(f + g) \geq \min\{v(f), v(g)\}$ still holds.*

Our first main result is the characterisation of non-zero quantum codes defined by group algebra elements.

Theorem 2. *Let $\Gamma = \text{Cay}(\mathbb{F}_2^r, S)$ be a Cayley graph over $\mathbb{F}_2^r$ with set of generators S . Let $\mathcal{C}$ be the quantum code on Γ and f its corresponding polynomial in the group algebra $\mathcal{A}_r$. The following holds:*

$$R(\mathcal{C}) = R(\ker f / \text{Im } f) > 0 \iff v(f) \geq 2.$$

Proof. The forward implication is Theorem 1 part (a).

For the converse, we prove a stronger statement that implies the claim by induction. In particular, let $f \in \mathcal{A}_r$ and write $f = x_r h + g$, $v(h) \geq 1, v(g) \geq 2$, $h, g \in \mathcal{A}_{r-1}$. Then $\ker f / \text{Im } f$ satisfies

$$\dim \frac{\ker f}{\text{Im } f} \geq \dim \frac{\ker g}{\text{Im } g}, \quad (1)$$

where $\ker f$ and $\text{Im } f$ are ideals of $\mathcal{A}_r$ and where $\ker g$ and $\text{Im } g$ are ideals of $\mathcal{A}_{r-1}$.

Consider $\alpha \in \ker g \setminus \text{Im } g$. Since αx_r clearly annihilates f ,

$$\text{either } \alpha x_r \in \ker f \setminus \text{Im } f \text{ or } \alpha x_r = (\lambda_0 + x_r \lambda_1)(x_r h + g) \in \text{Im } f.$$

The latter condition implies that $\lambda_0 g = 0$ and $\alpha = \lambda_0 h + \lambda_1 g$. Since $\lambda_0 h$ annihilates f and it is not in the image of g (otherwise α would also belong to it), $\lambda_0 h \in \ker f \setminus \text{Im } f$ (because polynomials not containing x_r in $\text{Im } f$ must be multiples of g). Moreover, $\lambda_0 h$ is in the same equivalence class of α (modulo $\text{Im } g$), and therefore, for different α , any two $\lambda_0 h$ cannot fall in the same equivalent class. Hence, each $\alpha \in \ker g / \text{Im } g$ defines either $\alpha x_r \in \ker f \setminus \text{Im } f$ or $\lambda_0 h \in \ker f \setminus \text{Im } f$, which proves the inequality (1).

In the base case of the induction $r = 1$, we have that the condition $v(f) \geq 2$ implies $f = 0$, hence the quotient space $\ker f / \text{Im } f$ equals the whole space $\mathcal{A}_1$ which is non-zero. Inequality (1) and the inductive hypotheses show that on any number of variables r the quotient space $\ker f / \text{Im } f$ is non-zero, meaning that the quantum code has non-zero dimension. $\square$

We have found it challenging to find lower bounds that grow with the number of variables r for the dimension of the quantum code. In particular, we believe that the dimension of the quantum code is bounded from below by $2^{\lceil r/2 \rceil}$, as this is the bound for quadratic forms proved in Section 4, and it is reasonable to believe that increasing valuations are related to higher dimensions of the quantum codes. In the following we provide partial results that support our hypotheses.

3 Separated Variables Formula

In this section we study polynomials of the type $H = f + g$ where f and g are defined on disjoint sets of variables. The main result is Theorem 3 below,

which gives a formula for the dimension of the ideal $\langle H \rangle$ in terms of the dimensions of the ideals $\langle f \rangle$ and $\langle g \rangle$. Additionally, the formula will sometimes allow us to give some insight into the structure of the quotient space $\ker H / \text{Im } H$, which can be thought of as a representation of the associated quantum code.

It will be useful to rename the ambient group algebra $\mathcal{A}$ as $\mathcal{A}_{[r]}$. For $J \subset [r]$, we denote by $\mathcal{A}_J$ the subalgebra of $\mathcal{A}$ generated by the functions 1 and $x_j, j \in J$. When the context is clear enough, we again just denote the ambient space $\mathcal{A}_{[r]} = \mathcal{A}_r$ as $\mathcal{A}$. If a function $f \in \mathcal{A}_{[r]}$ involves only variables x_j for $j \in J$ where $J \subset [r]$, we may consider f both as belonging to $\mathcal{A}_{[r]}$ and to the subring $\mathcal{A}_J$. It will be useful to consider the ideal generated by f both in the subring $\mathcal{A}_J$ and in the total ambient space $\mathcal{A}_{[r]}$. We shall therefore denote by $\dim_J f$ ($\dim_{[r]} f$) the dimension of the ideal $\langle f \rangle$ generated by f in $\mathcal{A}_J$ (in $\mathcal{A}_{[r]}$). Similarly, we write $\dim_J \ker f / \text{Im } f$ when f is considered in $\mathcal{A}_J$. By default, the absence of an index in $\dim$ means that the function is considered in the total ambient space $\mathcal{A}_{[r]}$.

Theorem 3. *Let $H = f + g \in \mathcal{A}_{[r]}$ such that $f \in \mathcal{A}_J$ and $g \in \mathcal{A}_{\bar{J}}$ and g contains no constant term 1. The following formula holds:*

$$\dim_{[r]} H = \dim_{[r]} f + \dim_{[r]} g - 2 \dim_J f \dim_{\bar{J}} g.$$

We omit the somewhat technical proof due to lack of space, and instead we focus on the important consequences. A first corollary of this proposition is an equivalent formula that provides the dimension of the quantum code defined by H and more interestingly, allows a generalisation to a sum of more than two polynomials in disjoint sets of variables.

Corollary 1. *Let $H = f + g \in \mathcal{A}_{[r]}$ such that $f \in \mathcal{A}_J$ and $g \in \mathcal{A}_{\bar{J}}$, and $v(g) \geq 1$. Then the dimension of the quantum code $\ker H / \text{Im } H$ satisfies*

$$\dim \frac{\ker H}{\text{Im } H} = \dim_J \frac{\ker f}{\text{Im } f} \cdot \dim_{\bar{J}} \frac{\ker g}{\text{Im } g},$$

More generally, for a partition $[r] = J_1 \sqcup \dots \sqcup J_k$, let $f_i \in \mathcal{A}_{J_i}$, $i = 1, \dots, k$ be polynomials with $v(f_i) \geq 1$, and define $H = \sum_{i=1}^k f_i$. Then the dimension of the quantum code $\ker H / \text{Im } H$ satisfies

$$\dim \frac{\ker H}{\text{Im } H} = \prod_{i=1}^k \dim_{J_i} \frac{\ker f_i}{\text{Im } f_i}.$$

Proof. Let $|J| = t$ and hence $|\bar{J}| = r - t$. The first statement is a direct

consequence of the previous proposition, as we have

$$\begin{aligned}
\dim \frac{\ker H}{\operatorname{Im} H} &= 2^r - 2 \dim_{[r]} H \\
&= 2^r - 2(\dim_{[r]} f + \dim_{[r]} g - 2 \dim_J f \dim_{\bar{J}} g) \\
&= (2^t - 2 \dim_J f) \cdot (2^{r-t} - 2 \dim_{\bar{J}} g) \\
&= \dim_J \frac{\ker f}{\operatorname{Im} f} \cdot \dim_{\bar{J}} \frac{\ker g}{\operatorname{Im} g}.
\end{aligned}$$

The generalisation to more than two polynomials is straightforward by induction on the number of monomials, applying the first part at each step. $\square$

As a second corollary we can describe the quantum code in case of sum of *completely separated* monomials, namely when all variables appear in at most one monomial.

Corollary 2. Let $f = \sum_{i=1}^k \prod_{j \in J_i} x_j \in \mathcal{A}_{[r]}$. In other words, $f = \sum m_i$ where m_i are monomials in disjoint sets of variables. Then the quantum code $\ker f / \operatorname{Im} f$ is the linear space generated by the set

$$S = \{n_1 n_2 \cdots n_k, n_i \text{ divides } m_i, 1 \neq n_i \neq m_i\} \otimes T,$$

where T is the linear space generated by 1 and all the monomials in the variables not appearing in f . In particular, the dimension of the quantum code is

$$\dim \frac{\ker f}{\operatorname{Im} f} = \prod_{i=1}^k (2^{k_i} - 2) \cdot 2^t,$$

where $k_i = |J_i|$ and $t = r - \sum k_i$ is the binary logarithm of the dimension of the space $T = \mathcal{A}_{[t]}$.

Clearly, this corollary is only meaningful in the case $|J_i| \geq 2$ for all $i = 1, \dots, k$, otherwise the quantum code is trivial.

Proof. The proof is a direct application of Theorem 3 and without loss of generality we can assume $t = 0$.

In fact, the elements in S are clearly in $\ker f$, but not in the image as elements in the image must contain at least a multiple of some m_i for some i . Moreover, different elements in S belong to different equivalence classes modulo $\operatorname{Im} f$, as no linear combination of them can contain a full monomial m_i . Finally, the number of such elements is exactly $\prod_{i=1}^k (2^{k_i} - 2)$, which equals the dimension of the quantum code by Corollary 1. $\square$

We are now in a position to construct the following example which shows that condition (b) in Theorem 1 is not an equivalence. The dimension computation is a consequence of the last corollary.

Example 1. Let $r = 9$ and consider the polynomial

$$f = x_1x_2x_3 + x_4x_5x_6 + x_7x_8x_9.$$

the rate of the quantum code $\ker f / \text{Im } f$ is

$$R(\ker f / \text{Im } f) = \frac{\dim(\ker f / \text{Im } f)}{2^9} = \left(\frac{2^3 - 2}{8} \right)^3 = \left(\frac{3}{4} \right)^3 < \frac{1}{2}.$$

In the last analysis, this example can be generalized to show that no positive constant lower bound of the rate of the quantum code for any fixed valuation exists.

In the following section we focus on the study of quantum codes defined by polynomials of valuation 2.

4 Quadratic Case

We begin this section by discussing the case of polynomials of valuation and degree exactly 2, i.e. quadratic forms. By the well known theory of quadratic forms (see [LN96]), any quadratic form q in the group algebra $\mathcal{A}$ can be written as

$$q = x'_1x'_2 + x'_3x'_4 + \cdots + x'_{2t-1}x'_{2t} \quad (2)$$

for $2t \leq r$ and linearly independent linear functions $x'_1, \dots, x'_{2t}$ of the variables $x_1, \dots, x_r$. The expression (2) is unique and referred to as the canonical form of q . Given a quadratic form q , t is often given the name *algebraic rank* (in symbols $\text{rk}(q)$). We call *full rank* the forms whose algebraic rank is the maximal value $\lfloor r/2 \rfloor$ in r variables.

The next result is the application of Corollary 2 to the quadratic case. It shows that the dimension of the principal ideal generated by a quadratic form q is uniquely determined by its rank $\text{rk}(q)$.

Proposition 1. Let q a quadratic form in r variables of algebraic rank $t \leq r/2$, then $\ker q / \text{Im } q$ is a monomial ideal generated by all the monomials where one variable of the couple (x_i, x_{i+1}) for every odd $i \leq 2t - 1$ appears exactly once. Consequently,

$$\dim(\ker q / \text{Im } q) = 2^{r-t}.$$

Notably, the dimension of the principal ideal generated by a full rank quadratic form is the maximal dimension of a principal ideal of $\mathcal{A}_r$ generated by a quadratic form.

In earlier studies (see [CDZ13]), polynomials f that are the sum of all monomials of degree higher than 1 are investigated, since they provide the first example of nontrivial quantum codes on graphs over binary vector

spaces. It turns out that the quadratic part of such polynomials is full rank and hence if one writes the quadratic part in canonical form, one gets $f = q + h$ where h is the higher valuation part. Here we provide a different and more precise analysis of the structure of the code, that hopefully can be employed in possible decoding algorithms. A fortiori, the formula for the rate of the code found in [CDZ13] will follow as special case.

Preliminarily, we focus on the case when f equals a quadratic form q . The following lemma is straightforward:

Lemma 1. For q a quadratic form on r variables (or in general for homogeneous polynomials), the ideal $\langle q \rangle$ can be uniquely decomposed into a direct sum of vector subspaces

$$\langle q \rangle = F_r \oplus F_{r-1} \oplus \cdots \oplus F_3 \oplus F_2$$

where the non-zero elements of F_i are homogeneous polynomials of degree i .

Next result emphasizes the fundamental role played by the canonical formula for quadratic forms.

Proposition 2. Let q be the full rank quadratic form in $r = 2k$ variables, then the dimensions of the spaces F_i defined in Lemma 1 can be computed recursively, starting from $k = 1$ where $\text{rk}(q) = 1$. We specify with $F_i^{(2j)}$ the direct summand F_i for a (full) rank j quadratic form in $2j$ variables. It holds:

$$\begin{cases} \dim F_i^{(2j)} = 2 \dim F_{i-1}^{(2j-2)} + \dim F_{i-2}^{(2j-2)} + \dim F_i^{(2j-2)} & i \neq j+1, \\ \dim F_i^{(2j)} = 2 \dim F_{i-1}^{(2j-2)} + \dim F_{i-2}^{(2j-2)} + \dim F_i^{(2j-2)} + 2^{j-1} & i = j+1; \end{cases}$$

where $F_i^{(j)} = \{0\}$ for $i \geq 2j$ and $i < 2$. Consequently, the dimension of the ideal generated by q is the sum of the values of $\dim F_i$ for $2 \leq i \leq n$. In symbols,

$$\dim \langle q \rangle = 2^{r-1} - 2^{\frac{r-2}{2}}.$$

Similar formulas hold for odd r .

The next theorem serves to compute the dimension of $\langle f \rangle$ for $f = q + h$, where q is full rank and h is any polynomial of valuation at least 3. In fact, it will be equal to the dimension of $\langle q \rangle$.

Theorem 4. Let $f \in \mathcal{A}$ be such that its quadratic part is of full rank. There is an isomorphism of algebras

$$\mathcal{A} = \frac{\mathbb{F}_2[x_1, \dots, x_r]}{\langle x_i^2 = 0 \ \forall i \leq r \rangle} \rightarrow \frac{\mathbb{F}_2[f_1, \dots, f_r]}{\langle f_i^2 = 0 \ \forall i \leq r \rangle}$$

$$x_i \mapsto f_i,$$

for $f_i = x_i + g_i$ with $v(g_i) \geq 2$.

In particular, if $f = q + h \in \mathcal{A}$ and q is full rank. Then the dimension of the quantum code $\mathcal{C}$ defined by f does not depend on h and it satisfies

$$\dim \mathcal{C} = \dim \ker q / \operatorname{Im} q = 2^{\lceil r/2 \rceil}.$$

The proof is based on the following *quasi-linear* change of variables, that incidentally states the formulas for the f_i 's.

Lemma 2. Let $f = q + \tilde{f}$ be an element of $\mathcal{A}$ such that $v(\tilde{f}) \geq 3$, q is the full rank quadratic form, and $\tilde{f}$ does not involve any variable not appearing in q . Then we can write $f = f_1 f_2 + \cdots + f_{2k-1} f_{2k}$ such that

$$\begin{cases} f_i = x_i + l_i(x_{i+2}, \dots, x_{2k}) + g_i(x_{i+2}, \dots, x_{2k})l_i(x_{i+2}, \dots, x_{2k}) & i \text{ odd,} \\ f_i = x_i + h_{i-1}(x_{i+1}, \dots, x_{2k}) + x_i g_{i-1}(x_{i+1}, \dots, x_{2k}) & i \text{ even;} \end{cases}$$

with $v(g_i) \geq 1$, $v(l_i) \geq 2$ and $v(h_i) \geq 2$ for all i .

Proof. The proof is by induction, once one can rearrange the terms in such a way that $f = f_1 f_2 + x_3 x_4 + \cdots + x_{2k-1} x_{2k} + \bar{f}(x_3, x_4, \dots, x_{2k})$, with $v(\bar{f}) \geq 3$, then continuing iteratively we obtain the result. We can write

$$f = x_1 x_2 + x_1 x_2 g_1 + x_1 h_1 + x_2 l_1 + x_3 x_4 + \cdots + x_{2k-1} x_{2k} + \bar{f}(x_3, x_4, \dots, x_{2k})$$

such that g_1, l_1, h_1 only contain the variables x_j , $j \geq 3$ and $v(\bar{f}) \geq 3$, $v(h_1), v(l_1) \geq 2$ and $v(g_1) \geq 1$. Then $f_1 f_2 = x_1 x_2 + x_2 l_1 + x_2 g_1 l_1 + x_1 h_1 + h_1 l_1 + g_1 l_1 h_1 + x_1 x_2 g_1 + x_2 g_1 l_1 + x_2 g_1^2 l_1 = x_1 x_2 + x_2 l_1 + x_1 h_1 + x_1 x_2 g_1 + (h_1 l_1 + h_1 l_1 g_1)$. Finally, we incorporate $h_1 l_1 + h_1 l_1 g_1$ in $\bar{f}$ (as they have valuation strictly greater than 2) without decreasing its valuation. $\square$

proof of the Theorem. Clearly, the minimal valuation part of f_i is simply x_i , so the map preserves the valuations. More precisely, $f_i f_j = 0 \iff x_i x_j = 0 \iff i = j$. This proves that the map is an injective morphism of algebras. Surjectivity is satisfied because a set of generators for the latter algebra is the image of a set of generators of the first algebra. $\square$

Remark 3. In the general case in which q is not of full rank we have the same isomorphism of algebras, but all variables successive to x_{2k} will be kept the same (that is $f_i = x_i$ for all $i > 2k$).

The combination of Theorem 4 and Theorem 3 allows to separate a quadratic part from the polynomial f , whose behaviour is well understood by Proposition 2, and study the remaining polynomial on fewer variables and valuation at least 3.

What is left unknown is the general behaviour of polynomials with valuation 3 and higher. It is not unrealistic to expect a lower bound on the dimension of the quantum codes above $2^{\lceil r/2 \rceil}$ as the valuation increases (for r variables), but we have found this issue to be challenging.

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